AutoML Experiment Report: Coronavirus_tweets_NLP

AutoHealth

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Abstract

This report details an automated machine learning (AutoML) experiment for multi-class sentiment classification of tweets related to the COVID-19 pandemic. The task involved classifying tweets into five sentiment categories: Extremely Negative, Negative, Neutral, Positive, and Extremely Positive. A comprehensive analysis revealed a dataset of 44,955 samples across training, validation, and test sets. A DeBERTa-v3-base model with handcrafted feature integration was employed, along with a Snapshot Ensemble strategy for robust uncertainty estimation. The final model achieved a test accuracy of 81.10% with an Expected Calibration Error (ECE) of 0.070, demonstrating well-calibrated predictions. Uncertainty quantification proved effective, with entropy serving as a reliable indicator of prediction error, enabling performance improvement through the rejection of high-uncertainty samples. This report provides a complete clinical-style analysis, including data insights, model development, performance evaluation, and actionable guidance for deployment.

1 Data Analysis

1.1 Dataset Overview

The dataset comprises 44,955 tweet samples organized into training (37,043 samples), validation (4,114 samples), and test (3,798 samples) sets. Each tweet is labeled with one of five sentiment categories: Extremely Negative, Negative, Neutral, Positive, and Extremely Positive. The data exhibits a moderate class imbalance, with Positive tweets being most frequent (10,280 in training) and Extremely Negative tweets being least frequent (4,933 in training).

1.2 Text Features and Key Challenges

Statistical analysis of text features revealed significant discriminative signals between sentiment classes:

- **Text Length:** Extremely Positive and Extremely Negative tweets are significantly longer (average character length >220) compared to Neutral tweets (average 168 characters).
- **Punctuation Patterns:** Neutral tweets have notably higher question mark usage (0.95 per tweet), while Extremely Positive tweets show elevated exclamation mark usage (0.28 per tweet).
- **Sentiment Vocabulary:** Extremely Positive tweets have the highest positive word ratio (0.0116), while Extremely Negative tweets have the highest negative word ratio (0.0050).
- **First Sentence Sentiment:** 24.94% of Extremely Positive tweets contain positive sentiment words in the first sentence.

Two critical data challenges were identified:

1. **Distribution Shift:** The test set shows lower bigram overlap (Jaccard similarity 0.3605) with training data and reduced cosine similarity in sentiment intensity patterns (0.8902 vs 0.9645 for train-validation), suggesting potential domain shift.
2. **Class Overlap:** The model faces challenges distinguishing between similar sentiment intensities (e.g., Negative vs Extremely Negative, Positive vs Extremely Positive).

1.3 Feature Engineering

Three handcrafted features were engineered to capture sentiment nuances:

- **Question Density:** Number of question marks divided by character length.
- **Exclamation Density:** Number of exclamation marks divided by character length.
- **First Sentence Sentiment Flag:** Boolean indicating presence of sentiment words in the first sentence.

These features were standardized using z-score normalization based on training set statistics.

2 Model Training

2.1 Data Preprocessing

Text preprocessing included: lowercase conversion, URL removal, @mention symbol removal (keeping username text), hashtag symbol removal (keeping tag text), and whitespace normalization.

2.2 Data Augmentation

A combined local augmentation strategy was applied to the training set to improve generalization:

- Synonym replacement using WordNet (max 3 words replaced, p=0.5).
- Random word swap (max 3 words swapped, p=0.5).
- Sentiment word insertion based on VADER lexicon (p=0.5).

The augmented training set expanded from 37,043 to 54,502 samples.

2.3 Model Architecture

The model employs a pre-trained **DeBERTa-v3-base** transformer with a custom classification head:

- **Base Model:** microsoft/deberta-v3-base produces 768-dimensional embeddings.
- **Feature Fusion:** Three handcrafted features (3 dimensions) are concatenated with the CLS token representation (768 dimensions) to form a 771-dimensional input.
- **Classification Head:** A linear layer maps 771 dimensions to 5 sentiment classes.

This lightweight fusion approach injects domain knowledge without excessive complexity.

2.4 Hyperparameters

Key hyperparameters were selected through a 4-trial hyperparameter search:

- **Optimizer:** AdamW with learning rate 2e-5 and weight decay 0.01.
- **Loss Function:** Cross-Entropy with label smoothing (0.1).
- **Batch Size:** 64 with gradient accumulation steps of 3 (effective batch size 192).
- **Learning Rate Schedule:** Linear warmup (10% steps) followed by cosine annealing.
- **Early Stopping:** Patience of 6 epochs on validation loss.

2.5 Training Process

Training was conducted with the following dynamics:

- Training loss decreased consistently from 1.64 to 0.58.
- Validation loss reached its minimum at epoch 4 (0.748) and plateaued thereafter.
- Best validation accuracy was 86.58% at epoch 6.

Figure 1: Training history showing loss and accuracy curves. The vertical dashed lines indicate snapshot collection points (epochs 1-4) used in the ensemble.

2.6 Model Performance

The final Snapshot Ensemble model was evaluated on the held-out test set. Temperature scaling was applied for calibration.

Table 1: Final Model Performance on Test Set

Metric	**Value**
Test Accuracy	0.8110
Expected Calibration Error (ECE)	0.070
Negative Log Likelihood (NLL)	0.569
Brier Score	0.288

3 Uncertainty Analysis

3.1 Uncertainty Estimation

Uncertainty was derived from the Snapshot Ensemble (4 models) with temperature scaling. Each snapshot's logits were divided by its learned temperature parameter before averaging probabilities. The predictive entropy of the averaged probabilities served as the uncertainty measure.

3.2 Model Calibration

The model demonstrated excellent calibration with an ECE of 0.070, indicating close agreement between predicted probabilities and actual correctness likelihood. Temperature scaling successfully improved calibration from the uncalibrated state.

3.3 Utility of Uncertainty

Despite good overall calibration, the model's uncertainty measures are informative:

- **Error Detection:** Higher entropy values are strongly associated with prediction errors. The entropy distribution for incorrect predictions peaks at higher values (0.8-1.0) compared to correct predictions (0.6-0.8).
- **Rejecting Uncertain Samples:** By rejecting samples with high uncertainty, accuracy improves significantly:
  - At 90% retention (threshold=1.146): accuracy = 84.09%
  - At 80% retention (threshold=1.008): accuracy = 86.03%

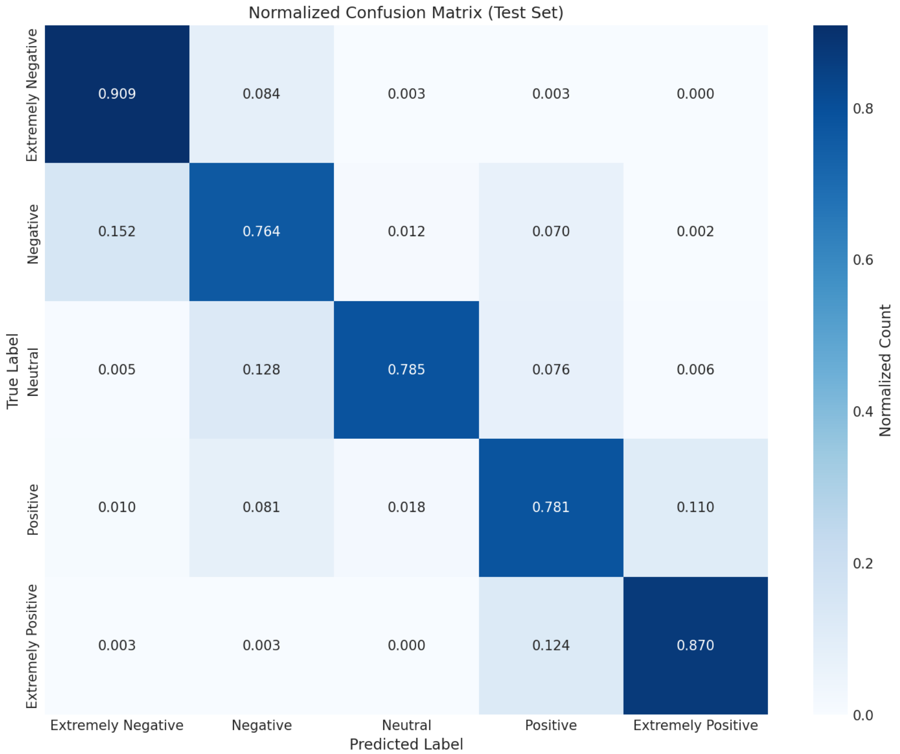

Figure 2: Normalized confusion matrix for the test set. The model shows strong performance on extreme sentiment classes (Extremely Negative: 0.909, Extremely Positive: 0.870) but experiences confusion between non-extreme categories.

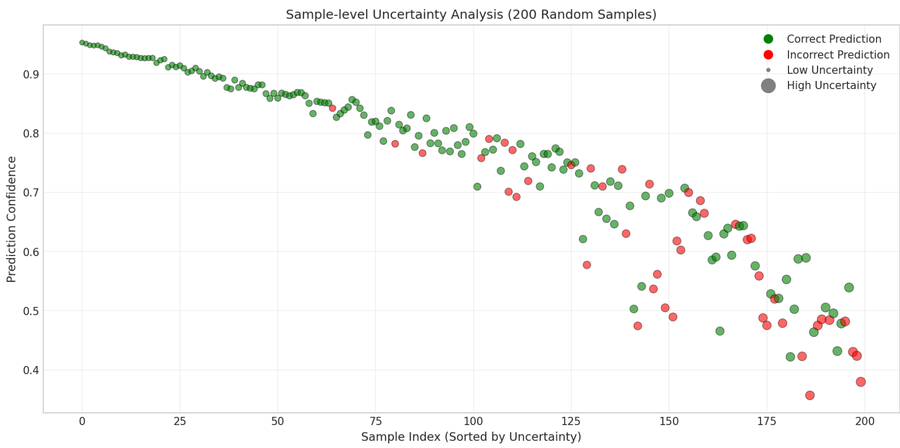

Figure 3: Sample-level uncertainty analysis for 200 random test samples. Green dots indicate correct predictions, red dots indicate errors. Larger dots represent higher entropy. Errors cluster in high-uncertainty regions.

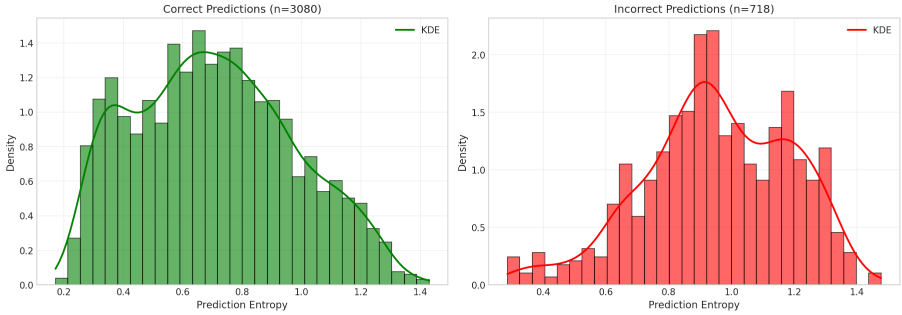

Figure 4: Prediction entropy distributions for correct (n=3080) and incorrect (n=718) predictions. Incorrect predictions exhibit higher entropy values, confirming the utility of entropy as an error indicator.

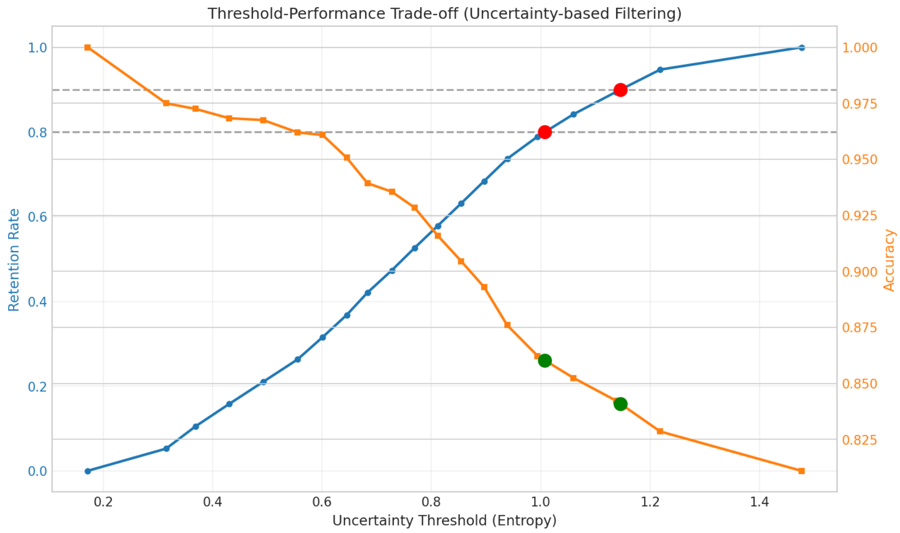

Figure 5: Performance trade-off when using entropy threshold for sample rejection. Higher thresholds (more aggressive rejection) lead to higher accuracy on retained samples but lower retention rates.

3.4 Feature Analysis for Class Confusion

Analysis of feature differences between correctly classified and confused samples reveals:

- **Negative vs Extremely Negative:** Confused samples show higher first-sentence sentiment flag values.
- **Positive vs Extremely Positive:** Confused samples show significantly higher exclamation density.

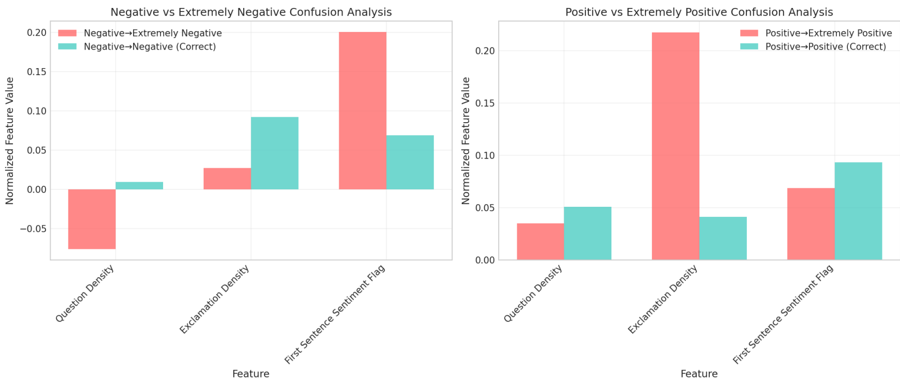

Figure 6: Feature comparison between correctly classified and confused samples for similar sentiment pairs. Key distinguishing features (exclamation density, first-sentence sentiment) show notable differences.

3.5 Distribution Shift Analysis

Comparison between validation and test set confidence distributions reveals some shift:

- Validation set concentrates more in high-confidence regions (0.8).
- Test set shows more density in mid-range confidence (0.4-0.6).

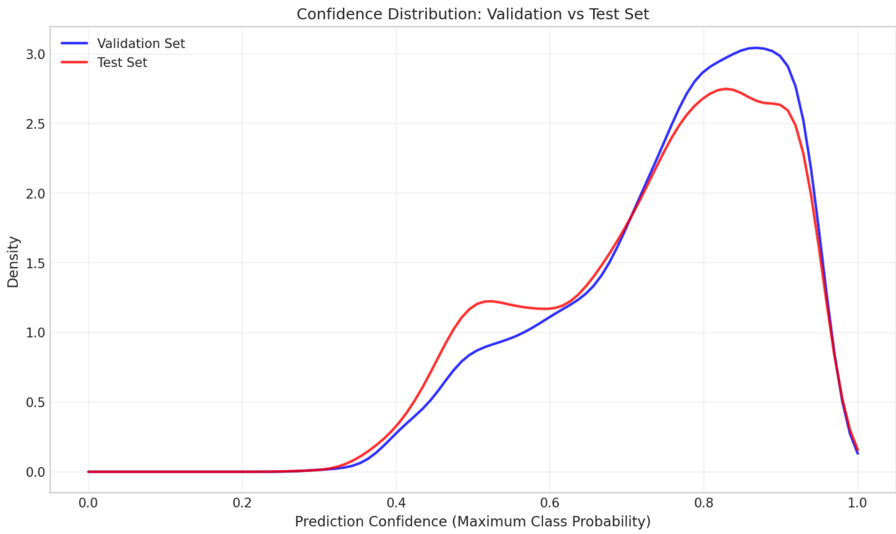

Figure 7: Confidence density distributions for validation (blue) and test (red) sets. The shift indicates potential domain differences between sets.

4 Conclusion

This AutoML experiment developed a sentiment classification model for COVID-19 related tweets, achieving 81.10% accuracy with well-calibrated predictions (ECE=0.070). The DeBERTa-v3-base model with handcrafted feature integration and Snapshot Ensemble strategy demonstrated effective performance on this multi-class task.

Key Strengths:

- Strong performance on extreme sentiment classes (recall > 0.87).
- Excellent model calibration (ECE=0.070).
- Effective uncertainty quantification for error detection.
- Uncertainty-based rejection can improve accuracy to 86% on high-confidence subset.

Key Limitations and Future Directions:

- **Class Confusion:** The model struggles to distinguish between similar sentiment intensities (e.g., Negative vs Extremely Negative). Future work could explore ordinal regression or focal loss to address this.
- **Distribution Shift:** The test set shows different confidence distributions than validation, suggesting potential domain shift. Future work should consider domain adaptation techniques.
- **Feature Engineering:** Additional features capturing emotional intensity may help distinguish between adjacent sentiment classes.

Actionable Guidance: The current model is suitable for sentiment analysis applications where high-confidence predictions can be trusted. For critical decisions, utilize the uncertainty estimates to flag low-confidence predictions for human review. The 80-86